FICTIONAL TEST PACKET. Not a real application.

Applicant's Name: Sam Okonkwo

STATEMENT OF PURPOSE

I am applying to Northwestern MSAI to learn the modeling I need for a project I have already started and cannot finish honestly. I work as an analyst in a city budget office. We publish spending tables. Residents ask which department grew, and I answer with spreadsheets. I want a model that can flag a line item that changed for a reason the table does not state, and I want to know when that model is wrong.

Since 2024 I have taken linear algebra, probability, and an introductory machine learning course through a university extension program, at night. I have not held a research job. My one project is a small classifier on ten years of public budget rows. It overfits. I can show the learning curve. I cannot yet fix it.

MSAI is the program I want because the coursework is the thing I am missing, and because I can keep the budget data as the project. I am not claiming a research record I do not have. I am applying so the first-year sequence, especially probability and machine learning, is the job I do next. If the committee wants a student who already ships models, I am not that student. If it wants a student with a defined public-data question and the start of the mathematics, I am.

EDUCATION

Undergraduate institution: DePaul University
Degree: B.A. Economics, 2023
Cumulative GPA: 3.45

RESEARCH

No research appointment. Independent study of public budget data, evenings, 2024 to the present, described under projects. I have not published.

PROJECTS

Budget-line classifier. A scikit-learn model on publicly posted city expenditure rows, ten years, one city. I held out the last year. Training accuracy was high and the holdout was near chance. I wrote a short memo about the leak, which was a feature that included the label. The code is in a public repository.

WORK EXPERIENCE

Budget analyst, City of Evanston finance department, July 2023 to the present. I prepare the quarterly spending tables. This is not a machine learning role. Extension coursework at night: linear algebra (A), probability (A-), introduction to machine learning (B+), 2024-2025.

HONORS AND AWARDS

Departmental honor in economics, DePaul, 2023.

LETTERS OF RECOMMENDATION

Letter 1. Director June Alvarez, City of Evanston. "Sam is the analyst I ask to explain a table to a resident. The writing is clear and the numbers are checked. Sam told me about the machine learning courses. I cannot evaluate that work. I can say the analyst work is reliable, and I support this application."

Letter 2. Dr. Robert Klein, university extension instructor. "Sam took my linear algebra and probability courses after work. The homework was complete and the exam grades were A and A-. I do not know Sam's research. I recommend Sam for further quantitative study."